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Counterexample to Sedykh and Shapiro conjecture

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Abstract

There is no finite constant C such that every closed convex curve γ in $\mathbb{R}^4$ satisfies $\mathrm{Vol}(\mathrm{EH}(\gamma)) \le C\ \mathrm{length}(\gamma)^4$. An explicit projective image of the normalized ellipse has length one and unbounded elliptic hull, of infinite volume. The same family, stopped short of the singular hyperplane, consists of length-one curves with compact elliptic hulls whose volumes are arbitrarily large.

1 The claim

A smooth closed curve γ in $\mathbb{R}^4$ is called *convex* when every affine hyperplane meets it in at most four points, counted with multiplicity. Multiplicity means the order of vanishing of $t \mapsto b \cdot \gamma(t) - p$ along the parametrization. At each point of the curves constructed below, the derivatives of orders $1, 2, 3$ are linearly independent, so there is a unique affine hyperplane of contact order 4. That hyperplane is supporting. The *elliptic hull* $\mathrm{EH}(\gamma)$ is the intersection of the corresponding closed half-spaces. It is closed and convex, and it contains the ordinary convex hull of the trace of γ.

Sedykh and Shapiro conjectured that, among closed convex curves of fixed length, the normalized ellipse uniquely maximizes the volume of every Young hull, and in particular the volume of the elliptic hull. A weaker form of the same idea is the existence of a universal isoperimetric inequality: a constant $C < \infty$, independent of the curve, such that

$$\mathrm{Vol}\big(\mathrm{EH}(\gamma)\big) \le C\ \mathrm{length}(\gamma)^4. \tag{1}$$

The ordinary convex hull does obey an inequality of this shape. The elliptic hull does not.

Theorem 1. *There is no finite constant C such that* (1) *holds for every closed convex curve in $\mathbb{R}^4$. The isoperimetric quotient $\mathrm{Vol}(\mathrm{EH}(\gamma))/\,\mathrm{length}(\gamma)^4$ is unbounded on that class, and it takes the value $+\infty$.*

The curves are projective images of the normalized ellipse

$$e(t) = \big(\sin t,\ \cos t,\ \tfrac{1}{2}\sin 2t,\ \tfrac{1}{2}\cos 2t\big), \qquad t \in \mathbb{R}/2\pi\mathbb{Z}. \tag{2}$$

Section 3 identifies $\mathrm{EH}(e)$ by explicit inequalities. Section 4 pushes those inequalities forward under the maps $x \mapsto x/(1 + \varepsilon x_3)$. Section 5 produces one curve of length 1 with infinite elliptic volume. Section 6 produces compact elliptic hulls of length-one curves with arbitrarily large volume. Section 7 records the contrasting bound for the ordinary convex hull.

Throughout, length means Euclidean length $\int_0^{2\pi} \|\gamma'(t)\|\, dt$, and volume means four-dimensional Lebesgue measure.

2 Trigonometric polynomials

Lemma 2. *Consider a trigonometric polynomial of order at most* 2*,*

$$f(t) = A + B\cos t + C\sin t + D\cos 2t + E\sin 2t.$$

The assignment $f \mapsto P$*, where* $z = e^{it}$ *and* $P(z) = z^2 f(t)$*, is a linear injection into the space of algebraic polynomials of degree at most* 4*. Consequently, either* f *vanishes identically or* f *has at most four zeros on* $\mathbb{R}/2\pi\mathbb{Z}$*, counted with multiplicity. The five functions* $1, \cos t, \sin t, \cos 2t, \sin 2t$ *are linearly independent.*

Proof. With $z = e^{it}$,

$$z^2\cos t = \tfrac{1}{2}(z^3 + z), \qquad z^2\sin t = \tfrac{1}{2i}(z^3 - z),$$
$$z^2\cos 2t = \tfrac{1}{2}(z^4 + 1), \qquad z^2\sin 2t = \tfrac{1}{2i}(z^4 - 1).$$

Hence

$$P(z) = \Big(\frac{D}{2} + \frac{E}{2i}\Big)z^4 + \Big(\frac{B}{2} + \frac{C}{2i}\Big)z^3 + Az^2 + \Big(\frac{B}{2} - \frac{C}{2i}\Big)z + \Big(\frac{D}{2} - \frac{E}{2i}\Big).$$

This is a polynomial of degree at most 4. If P is the zero polynomial, the coefficients of z^4 and z^0 give $D = E = 0$, those of z^3 and z^1 give $B = C = 0$, and the coefficient of z^2 gives $A = 0$. Thus $f \mapsto P$ is injective, and the five trigonometric functions are independent.

The function e^{2it} never vanishes, so f and $t \mapsto P(e^{it})$ have zeros of the same order. It remains to compare the order of $P(e^{it})$ with the order of P at e^{it_0}. If P is identically zero there is nothing to prove. Otherwise factor $P(z) = (z - z_0)^k R(z)$ with $R(z_0) \neq 0$ and $z_0 = e^{it_0}$. Then

$$P(e^{it}) = \big(e^{it} - e^{it_0}\big)^k R(e^{it}) = e^{ikt_0}\big(e^{i(t-t_0)} - 1\big)^k R(e^{it}).$$

The function $h(\theta) = e^{i\theta} - 1$ has a zero of order exactly 1 at $\theta = 0$, because $h(0) = 0$ and $h'(0) = i \neq 0$. If a smooth function has a zero of order m and another has a zero of order n, their product has a zero of order $m + n$: in Leibniz's rule for the derivative of order $m + n$, every term except the one that differentiates the first factor m times and the second n times vanishes, and that term is nonzero. Iterating, h^k has a zero of order k. The factor $R(e^{it_0})$ is nonzero, so $P(e^{it})$ has order exactly k at t_0. Distinct points of $\mathbb{R}/2\pi\mathbb{Z}$ give distinct values of e^{it}, so the sum of the orders of the zeros of f equals the sum of the orders of the corresponding roots of P, which is at most $\deg P \leq 4$. □

3 The normalized ellipse

Lemma 3. *The curve* e *is convex, and* $\operatorname{span}\{e'(t), e''(t), e'''(t)\}$ *is three-dimensional at every* t*.*

Proof. The function $a \cdot e(t) - c$ is a trigonometric polynomial of order at most 2. It vanishes identically only if $a = 0$ and $c = 0$, by the linear independence in Lemma 2 and the expansion

$$a \cdot e(t) = a_1\sin t + a_2\cos t + \tfrac{1}{2}a_3\sin 2t + \tfrac{1}{2}a_4\cos 2t.$$

Lemma 2 therefore limits every affine hyperplane to at most four intersections with e.

Let R_θ be the plane rotation

$$R_\theta \begin{pmatrix} u \\ v \end{pmatrix} = \begin{pmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{pmatrix} \begin{pmatrix} u \\ v \end{pmatrix}.$$

Angle addition gives $R_{\theta+\varphi} = R_\theta R_\varphi$. Define the block-diagonal rotation $A(t)(x', x'') = (R_t x', R_{2t} x'')$ on $\mathbb{R}^2 \times \mathbb{R}^2$. Then $A(t+s) = A(t)A(s)$ and $A(t)^{-1} = A(-t)$. Starting from $e(0) = (0, 1, 0, 1/2)$,

$$A(t)e(0) = \big(\sin t, \cos t, \tfrac{1}{2}\sin 2t, \tfrac{1}{2}\cos 2t\big) = e(t).$$

The identity $e(t+s) = A(t)e(s)$ follows, and differentiation in s yields $e^{(k)}(t) = A(t)e^{(k)}(0)$ for every $k \geq 1$. Since $A(t)$ is invertible, the rank of $\{e', e'', e'''\}$ is independent of t. At $t = 0$,

$$e'(0) = (1, 0, 1, 0), \qquad e''(0) = (0, -1, 0, -2), \qquad e'''(0) = (-1, 0, -4, 0).$$

If $ae'(0) + be''(0) + ce'''(0) = 0$, the second coordinate gives $-b = 0$, and the first and third coordinates then give $a - c = 0$ and $a - 4c = 0$, so $a = b = c = 0$. □

Define

$$n(s) = \big(6\sin s,\ 6\cos s,\ -3\sin 2s,\ -3\cos 2s\big). \tag{3}$$

Lemma 4. *For every real s and u,*

$$n(s) \cdot e(s+u) - \tfrac{9}{2} = -3(\cos u - 1)^2. \tag{4}$$

The right-hand side has a zero of order exactly 4 at $u = 0$. In particular $n(s) \cdot e(s) = 9/2$, the vector $n(s)$ is orthogonal to $e'(s)$, $e''(s)$ and $e'''(s)$, and $\|n(s)\|^2 = 45$.

Proof. Expand the inner product by the cosine subtraction formula:

$$\begin{aligned} n(s) \cdot e(s+u) &= 6\big(\sin s \sin(s+u) + \cos s \cos(s+u)\big) \\ &\quad - \tfrac{3}{2}\big(\sin 2s \sin 2(s+u) + \cos 2s \cos 2(s+u)\big) \\ &= 6\cos u - \tfrac{3}{2}\cos 2u. \end{aligned}$$

Substitute $\cos 2u = 2\cos^2 u - 1$:

$$\begin{aligned} 6\cos u - \tfrac{3}{2}\cos 2u - \tfrac{9}{2} &= 6\cos u - 3\cos^2 u + \tfrac{3}{2} - \tfrac{9}{2} \\ &= -3\big(\cos^2 u - 2\cos u + 1\big) = -3(\cos u - 1)^2. \end{aligned}$$

Let $w(u) = \cos u - 1$. Then $w(0) = 0$, $w'(0) = -\sin 0 = 0$ and $w''(0) = -\cos 0 = -1$, so w has a zero of order 2 at the origin. Leibniz's rule gives

$$(w^2)^{(m)}(0) = \sum_{j=0}^{m} \binom{m}{j} w^{(j)}(0)\, w^{(m-j)}(0).$$

If $m < 4$, then in every summand at least one of j and $m - j$ is at most 1, so the summand vanishes. If $m = 4$, the only surviving summand is $j = 2$, and

$$\binom{4}{2} w''(0) w''(0) = 6(-1)(-1) = 6 \neq 0.$$

Thus w^2 has a zero of order exactly 4, and so does $-3w^2$.

Setting $u = 0$ gives $n(s) \cdot e(s) = 9/2$. Differentiating (4) once, twice and three times in u and then setting $u = 0$ gives the three orthogonality relations. Finally

$$\|n(s)\|^2 = 36(\sin^2 s + \cos^2 s) + 9(\sin^2 2s + \cos^2 2s) = 45.$$

□

Proposition 5. *The elliptic hull of the normalized ellipse is the compact convex set*

$$\mathrm{EH}(e) = \big\{x \in \mathbb{R}^4 : n(s) \cdot x \le \tfrac{9}{2} \text{ for every } s\big\}. \tag{5}$$

On this set, $|x_3| \le 3/2$ *and* $|x_4| \le 3/2$. *The closed Euclidean ball of radius* $3/(2\sqrt{5})$ *about the origin is contained in* $\mathrm{EH}(e)$, *and so is the point*

$$p_* = \big(0, 0, -\tfrac{3}{2}, 0\big). \tag{6}$$

The bound $x_3 \ge -3/2$ *is attained at* p_*.

Proof. Fix s. By Lemmas 3 and 4, the vectors $e'(s), e''(s), e'''(s)$ form a basis of the hyperplane orthogonal to the nonzero vector $n(s)$. The unique affine hyperplane through $e(s)$ containing that 3-space is $\{x : n(s) \cdot x = 9/2\}$. Lemma 4 says that $t \mapsto n(s) \cdot e(t) - 9/2$ has a zero of order exactly 4 at $t = s$ and no other zeros, and that the function is nonpositive. Thus these are precisely the osculating support inequalities, and their intersection is $\mathrm{EH}(e)$.

The third coordinate of $n(s)$ is $-3\sin 2s$, so

$$n(s) \cdot p_* = (-3\sin 2s)\big(-\tfrac{3}{2}\big) = \tfrac{9}{2}\sin 2s \le \tfrac{9}{2}.$$

Hence $p_* \in \mathrm{EH}(e)$. The origin satisfies every inequality strictly. If $\|x\| \le 3/(2\sqrt{5})$, then

$$n(s) \cdot x \le \|n(s)\|\,\|x\| \le \sqrt{45} \cdot \frac{3}{2\sqrt{5}} = 3\sqrt{5} \cdot \frac{3}{2\sqrt{5}} = \tfrac{9}{2},$$

so the closed ball lies in $\mathrm{EH}(e)$.

The same normal, evaluated at explicit angles, controls the third and fourth coordinates. One has

$$\begin{aligned} n(\pi/4) &= (3\sqrt{2}, 3\sqrt{2}, -3, 0), & n(5\pi/4) &= (-3\sqrt{2}, -3\sqrt{2}, -3, 0), \\ n(3\pi/4) &= (3\sqrt{2}, -3\sqrt{2}, 3, 0), & n(7\pi/4) &= (-3\sqrt{2}, 3\sqrt{2}, 3, 0), \\ n(0) &= (0, 6, 0, -3), & n(\pi) &= (0, -6, 0, -3), \\ n(\pi/2) &= (6, 0, 0, 3), & n(3\pi/2) &= (-6, 0, 0, 3). \end{aligned}$$

The average of the first pair is $(0, 0, -1, 0)$. Each half-space contributes at most $9/2$, so

$$-x_3 = \tfrac{1}{6}n(\pi/4) \cdot x + \tfrac{1}{6}n(5\pi/4) \cdot x \le \tfrac{3}{2}.$$

The average of the second pair is $(0, 0, 1, 0)$, and the same estimate yields $x_3 \le 3/2$. The average of $n(0)$ and $n(\pi)$ is $(0, 0, 0, -1)$, so $x_4 \ge -3/2$. The average of $n(\pi/2)$ and $n(3\pi/2)$ is $(0, 0, 0, 1)$, so $x_4 \le 3/2$.

At p_* one has $x_3 = -3/2$, so the lower bound on x_3 is sharp. Both $n(\pi/4)$ and $n(5\pi/4)$ give the value $9/2$ there, since $\sin(\pi/2) = \sin(5\pi/2) = 1$.

The set (5) is closed. To see that it is bounded, write

$$n(s) \cdot x = 6(x_1 \sin s + x_2 \cos s) - 3(x_3 \sin 2s + x_4 \cos 2s)$$

and set $A = \sqrt{x_1^2 + x_2^2}$, $B = \sqrt{x_3^2 + x_4^2}$. As s runs through a full period, so does $2s$, and

$$\max(P\sin\theta + Q\cos\theta) = \sqrt{P^2 + Q^2}.$$

Thus the first summand has maximum $6A$ and the second has minimum $-3B$. Evaluating the sum at a maximum point of the first summand shows that its maximum is at least $6A - 3B$. On $\mathrm{EH}(e)$ that maximum is at most $9/2$. The coordinate bounds give $B \le \frac{3}{2}\sqrt{2}$, so

$$6A \le \tfrac{9}{2} + 3 \cdot \tfrac{3}{2}\sqrt{2} = \tfrac{9}{2}(1 + \sqrt{2}), \qquad A \le \tfrac{3}{4}(1 + \sqrt{2}).$$

Therefore $\mathrm{EH}(e)$ is closed and bounded, hence compact. □

4 The projective family

For $\varepsilon \in [0, 2/3]$ define

$$\rho_\varepsilon(t) = 1 + \frac{\varepsilon}{2}\sin 2t, \qquad \gamma_\varepsilon(t) = \frac{e(t)}{\rho_\varepsilon(t)}, \qquad \phi_\varepsilon(x) = \frac{x}{1+\varepsilon x_3}. \tag{7}$$

The scalar ρ_ε satisfies $\rho_\varepsilon(t) \geq 1 - \varepsilon/2 \geq 2/3 > 0$, so γ_ε is a smooth 2π-periodic curve, and $\gamma_\varepsilon = \phi_\varepsilon \circ e$. On the half-space $\{y : 1 - \varepsilon y_3 > 0\}$ the inverse map is $\psi_\varepsilon(y) = y/(1 - \varepsilon y_3)$. Write

$$D_\varepsilon = \{x : 1 + \varepsilon x_3 > 0\}.$$

Lemma 6. *The map ϕ_ε is a bijection of D_ε onto $\{y : 1 - \varepsilon y_3 > 0\}$, with inverse ψ_ε.*

Proof. If $x \in D_\varepsilon$ and $y = \phi_\varepsilon(x)$, then $y_3 = x_3/(1 + \varepsilon x_3)$, so

$$1 - \varepsilon y_3 = \frac{1}{1+\varepsilon x_3} > 0$$

and $\psi_\varepsilon(y) = x$. Conversely, if $1 - \varepsilon y_3 > 0$ and $x = \psi_\varepsilon(y)$, the same algebra gives $x \in D_\varepsilon$ and $\phi_\varepsilon(x) = y$. □

Lemma 7. *For every $\varepsilon \in [0, 2/3]$ and every t,*

$$e(t) \cdot e'(t) = 0, \qquad \|e(t)\|^2 = \tfrac{5}{4}, \qquad \|e'(t)\|^2 = 2,$$

and, writing $\rho = \rho_\varepsilon$,

$$\|\gamma_\varepsilon'(t)\| = \frac{\sqrt{2\rho(t)^2 + \frac{5}{4}\rho'(t)^2}}{\rho(t)^2}. \tag{8}$$

In particular $\|\gamma_\varepsilon'(t)\| \geq \sqrt{2}/\rho(t) > 0$, and

$$\|\gamma_\varepsilon'(t)\| \leq \frac{3\sqrt{37}}{4}, \qquad 0 < \operatorname{length}(\gamma_\varepsilon) \leq \frac{3\pi\sqrt{37}}{2}.$$

The length of e itself is $2\pi\sqrt{2}$.

Proof. Differentiating (2) gives $e' = (\cos t, -\sin t, \cos 2t, -\sin 2t)$. Then

$$\begin{aligned} e \cdot e' &= \sin t \cos t - \cos t \sin t + \tfrac{1}{2}\sin 2t \cos 2t - \tfrac{1}{2}\cos 2t \sin 2t = 0, \\ \|e\|^2 &= 1 + \tfrac{1}{4} = 5/4, \\ \|e'\|^2 &= 1 + 1 = 2. \end{aligned}$$

Hence $\operatorname{length}(e) = \int_0^{2\pi} \sqrt{2}\, dt = 2\pi\sqrt{2}$. With $\rho' = \varepsilon \cos 2t$,

$$\gamma_\varepsilon' = \frac{\rho e' - \rho' e}{\rho^2},$$

and the cross term drops out, so

$$\|\rho e' - \rho' e\|^2 = 2\rho^2 + \tfrac{5}{4}(\rho')^2.$$

Dividing by ρ^4 produces (8). The numerator is at least $\rho\sqrt{2}$, so the speed is at least $\sqrt{2}/\rho > 0$.

For the upper bound, $\varepsilon \le 2/3$ gives $2/3 \le \rho \le 4/3$ and $|\rho'| \le \varepsilon \le 2/3$. The numerator of (8) is therefore at most

$$\sqrt{2\Big(\frac{4}{3}\Big)^2 + \frac{5}{4}\Big(\frac{2}{3}\Big)^2} = \sqrt{\frac{32}{9} + \frac{5}{9}} = \frac{\sqrt{37}}{3},$$

while the denominator is at least $(2/3)^2 = 4/9$. Hence

$$\|\gamma_\varepsilon'\| \le \frac{\sqrt{37}}{3} \cdot \frac{9}{4} = \frac{3\sqrt{37}}{4}.$$

Integrating over an interval of length 2π gives the length bound. The integrand is continuous and positive, so the length is positive and finite. □

Proposition 8. *For each $\varepsilon \in [0, 2/3]$ the curve γ_ε is convex. At every parameter the vectors $\gamma_\varepsilon', \gamma_\varepsilon'', \gamma_\varepsilon'''$ are linearly independent, and the osculating hyperplane of γ_ε at $\gamma_\varepsilon(s)$ is*

$$n(s) \cdot y + \tfrac{9}{2}\varepsilon\, y_3 = \tfrac{9}{2}. \tag{9}$$

The curve lies in the closed half-space in which the left-hand side is at most $9/2$.

Proof. An affine equation $b \cdot y = p$ pulls back along $\gamma_\varepsilon = e/\rho_\varepsilon$ to the trigonometric equation

$$b \cdot e(t) - p\rho_\varepsilon(t) = 0.$$

Expanding,

$$b \cdot e(t) - p\rho_\varepsilon(t) = b_1 \sin t + b_2 \cos t + \Big(\frac{b_3}{2} - \frac{p\varepsilon}{2}\Big)\sin 2t + \frac{b_4}{2}\cos 2t - p.$$

This lies in the space of Lemma 2. If it vanished identically, the constant term would force $p = 0$ and the remaining coefficients would force $b = 0$, which is not the equation of a hyperplane. Lemma 2 therefore yields at most four zeros. Division by the nonvanishing function ρ_ε does not change orders, so γ_ε is convex.

Let $g_s(t)$ be the left-hand side of (9), evaluated on γ_ε, minus $9/2$. Substituting $y = e/\rho_\varepsilon$ and $\rho_\varepsilon = 1 + \varepsilon e_3$ gives the numerator

$$n(s) \cdot e + \tfrac{9}{2}\varepsilon e_3 - \tfrac{9}{2}\rho_\varepsilon = n(s) \cdot e + \tfrac{9}{2}\varepsilon e_3 - \tfrac{9}{2} - \tfrac{9}{2}\varepsilon e_3 = n(s) \cdot e - \tfrac{9}{2},$$

and therefore, by Lemma 4,

$$g_s(t) = \frac{-3\big(\cos(t-s) - 1\big)^2}{\rho_\varepsilon(t)}.$$

The denominator does not vanish, so g_s is nonpositive and has a zero of order exactly 4 at $t = s$. Equation (9) is consequently a supporting hyperplane of contact order 4.

To see that it is the only such hyperplane, let T be the space of trigonometric polynomials of order at most 2. Evaluation of the 4-jet $(f(t_0), f'(t_0), f''(t_0), f'''(t_0))$ is a linear map $T \to \mathbb{R}^4$. Translation $f(t) \mapsto f(t + t_0)$ is an automorphism of T, and it identifies the 4-jet at t_0 with the 4-jet at 0. It is therefore enough to check surjectivity at $t_0 = 0$. For

$$f(t) = a\cos t + b\sin t + c\cos 2t + d\sin 2t$$

one has

$$f(0) = a + c, \qquad f'(0) = b + 2d,$$
$$f''(0) = -a - 4c, \qquad f'''(0) = -b - 8d.$$

The blocks $(a,c) \mapsto (a+c, -a-4c)$ and $(b,d) \mapsto (b+2d, -b-8d)$ have matrices

$$\begin{pmatrix} 1 & 1 \\ -1 & -4 \end{pmatrix}, \qquad \begin{pmatrix} 1 & 2 \\ -1 & -8 \end{pmatrix},$$

with determinants -3 and -6. Both are invertible, so these four functions already have surjective 4-jet. The domain T is five-dimensional, and the jet map has rank 4, so its kernel is one-dimensional. By Lemma 4 the function $p(t) = n(t_0) \cdot e(t) - 9/2$ is a nonzero element of that kernel. It therefore spans the kernel.

Now suppose $\alpha \cdot \gamma_\varepsilon'(t_0) = \alpha \cdot \gamma_\varepsilon''(t_0) = \alpha \cdot \gamma_\varepsilon'''(t_0) = 0$, and set $c = \alpha \cdot \gamma_\varepsilon(t_0)$. The function $\alpha \cdot \gamma_\varepsilon - c$ has a zero of order at least 4 at t_0. Multiplying by the nonvanishing function ρ_ε shows that

$$q(t) = \alpha \cdot e(t) - c\rho_\varepsilon(t)$$

has a zero of order at least 4 at t_0. Also $q \in T$, so $q = \lambda p$ for some scalar λ.

The average value of $(\cos u - 1)^2 = \cos^2 u - 2\cos u + 1$ is $\frac{1}{2} + 1 = \frac{3}{2}$. Averaging (4) therefore gives

$$\frac{1}{2\pi} \int_0^{2\pi} n(t_0) \cdot e(t)\, dt - \tfrac{9}{2} = -3 \cdot \tfrac{3}{2} = -\tfrac{9}{2},$$

so $t \mapsto n(t_0) \cdot e(t)$ has mean zero. The constant term of $\alpha \cdot e$ is 0, and the constant term of ρ_ε is 1. Comparing constant terms in $q = \lambda p$ yields $c = (9/2)\lambda$. Substituting this value and $\rho_\varepsilon = 1 + (\varepsilon/2)\sin 2t$ gives

$$\alpha \cdot e(t) = \lambda\, n(t_0) \cdot e(t) + \lambda \cdot \frac{9\varepsilon}{4} \sin 2t.$$

The right-hand side is determined by λ. The four coordinate functions of e are linearly independent, so α itself is determined by λ. The annihilator of $\operatorname{span}\{\gamma_\varepsilon'(t_0), \gamma_\varepsilon''(t_0), \gamma_\varepsilon'''(t_0)\}$ is therefore at most one-dimensional.

It is exactly one-dimensional because the conormal of (9) lies in it and is nonzero: the computation of g_s shows that this conormal annihilates the first three derivatives, and its first two components are $6\sin s$ and $6\cos s$, which do not vanish simultaneously. The three derivatives are linearly independent, and the hyperplane of contact order 4 is unique. □

Proposition 9. *Let $\varepsilon \in [0, 2/3]$. A point $x \in \mathrm{EH}(e) \cap D_\varepsilon$ has image $\phi_\varepsilon(x) \in \mathrm{EH}(\gamma_\varepsilon)$. Conversely, every point of $\mathrm{EH}(\gamma_\varepsilon)$ lies in $\{y : 1 - \varepsilon y_3 > 0\}$, and*

$$\mathrm{EH}(\gamma_\varepsilon) = \phi_\varepsilon\big(\mathrm{EH}(e) \cap D_\varepsilon\big).$$

If $0 \le \varepsilon < 2/3$, then $\mathrm{EH}(e) \subset D_\varepsilon$ and $\mathrm{EH}(\gamma_\varepsilon) = \phi_\varepsilon(\mathrm{EH}(e))$ is compact.

Proof. Take $x \in D_\varepsilon$ and set $y = \phi_\varepsilon(x)$, so $x = \psi_\varepsilon(y)$ and $1 - \varepsilon y_3 > 0$ by Lemma 6. For each s,

$$\begin{aligned} n(s) \cdot x \le \tfrac{9}{2} &\iff \frac{n(s) \cdot y}{1 - \varepsilon y_3} \le \tfrac{9}{2} \\ &\iff n(s) \cdot y + \tfrac{9}{2}\varepsilon\, y_3 \le \tfrac{9}{2}. \end{aligned}$$

The division preserves the inequality because the denominator is positive. By Proposition 8, the last family is exactly the family of osculating inequalities of γ_ε. Thus $x \in \mathrm{EH}(e) \cap D_\varepsilon$ implies $y \in \mathrm{EH}(\gamma_\varepsilon)$, which is the forward inclusion

$$\phi_\varepsilon\big(\mathrm{EH}(e) \cap D_\varepsilon\big) \subset \mathrm{EH}(\gamma_\varepsilon).$$

For the reverse inclusion, take $y \in \mathrm{EH}(\gamma_\varepsilon)$. Proposition 8 gives, for every s,

$$6(y_1 \sin s + y_2 \cos s) - 3(y_3 \sin 2s + y_4 \cos 2s) + \tfrac{9}{2}\varepsilon y_3 \le \tfrac{9}{2}. \tag{10}$$

Integrate over $[0, 2\pi]$. Every nonconstant Fourier mode integrates to zero, and one obtains $\frac{9}{2}\varepsilon y_3 \le \frac{9}{2}$. If $\varepsilon = 0$, then ϕ_0 is the identity and $D_0 = \mathbb{R}^4$, so the two sets agree and there is nothing further to prove. If $\varepsilon > 0$, then $y_3 \le 1/\varepsilon$.

Equality $y_3 = 1/\varepsilon$ is impossible. It would force the integrand of the averaged inequality to equal the constant $9/2$ everywhere, because a continuous function bounded above by its mean is constant. The nonconstant part of (10) would vanish identically. Lemma 2 would give $y_1 = y_2 = y_3 = y_4 = 0$, contradicting $y_3 = 1/\varepsilon$. Therefore $y_3 < 1/\varepsilon$, the point y lies in the domain of ψ_ε, and the equivalence of the first paragraph puts $\psi_\varepsilon(y)$ in $\mathrm{EH}(e) \cap D_\varepsilon$. Hence y lies in $\phi_\varepsilon(\mathrm{EH}(e) \cap D_\varepsilon)$.

Finally, if $\varepsilon < 2/3$, Proposition 5 gives $x_3 \ge -3/2$ on the compact set $\mathrm{EH}(e)$, so

$$1 + \varepsilon x_3 \ge 1 - \tfrac{3}{2}\varepsilon > 0$$

and $\mathrm{EH}(e) \subset D_\varepsilon$. The continuous image $\phi_\varepsilon(\mathrm{EH}(e))$ is compact, and the two inclusions identify it with $\mathrm{EH}(\gamma_\varepsilon)$. □

Positive homotheties preserve the whole construction.

Lemma 10. *Let γ be any of the curves γ_ε, and let $c > 0$. The curve $\eta(t) = c\gamma(t)$ is closed and convex,* $\mathrm{length}(\eta) = c\,\mathrm{length}(\gamma)$, *and* $\mathrm{EH}(\eta) = c\ \mathrm{EH}(\gamma)$. *In particular,* $\mathrm{EH}(\eta)$ *is compact, unbounded, or of infinite volume precisely when* $\mathrm{EH}(\gamma)$ *is.*

Proof. If $b \cdot x = p$ is an affine hyperplane, then $b \cdot \eta(t) - cp = c\big(b \cdot \gamma(t) - p\big)$. Since $c \neq 0$, the two functions have the same zeros with the same orders. Thus η is convex. The speed of η is c times the speed of γ, so the lengths stand in the same ratio.

The derivatives are $\eta^{(k)} = c\gamma^{(k)}$. They span the same three-dimensional space as the derivatives of γ, by Proposition 8. If $b \cdot x = p$ is the osculating hyperplane of γ at $\gamma(t)$, then b annihilates that space and $b \cdot \gamma(t) = p$, so $b \cdot \eta(t) = cp$ and $\{x : b \cdot x = cp\}$ is the osculating hyperplane of η. For $c > 0$, the curve γ lies in $b \cdot x \le p$ if and only if η lies in $b \cdot x \le cp$. A point z satisfies every osculating inequality of η if and only if z/c satisfies every osculating inequality of γ. Therefore $\mathrm{EH}(\eta) = c\ \mathrm{EH}(\gamma)$.

Distances scale by c and four-dimensional volumes by c^4. A set of infinite measure is sent to a set of infinite measure, because it contains simplices of arbitrarily large finite volume and those volumes are multiplied by the positive constant c^4. □

5 A curve of length one with infinite elliptic volume

Theorem 11. *There exists a closed convex curve in $\mathbb{R}^4$ of length 1 whose elliptic hull is unbounded and has infinite Lebesgue measure.*

Proof. Take $\varepsilon = 2/3$, and write $\gamma_* = \gamma_{2/3}$ and $\phi = \phi_{2/3}$. Lemma 7 gives

$$0 < \mathrm{length}(\gamma_*) \le \frac{3\pi\sqrt{37}}{2},$$

and Proposition 8 says that γ_* is a closed convex curve. For $a \in [0, 1)$ the point

$$x_a = \big(0, 0, -\tfrac{3a}{2}, 0\big) = (1 - a) \cdot 0 + a\, p_*$$

lies on the segment joining the origin to p_*. Both endpoints lie in the convex set $\mathrm{EH}(e)$, so $x_a \in \mathrm{EH}(e)$. Its denominator is

$$1 + \tfrac{2}{3}\Big(-\frac{3a}{2}\Big) = 1 - a > 0,$$

and

$$\phi(x_a) = \Big(0, 0, -\frac{3a}{2(1-a)}, 0\Big).$$

Proposition 9 places $\phi(x_a)$ in $\mathrm{EH}(\gamma_*)$. The distance from the origin to this point equals $3a/(2(1-a))$, which tends to infinity as $a \uparrow 1$. The origin is $\phi(0) \in \mathrm{EH}(\gamma_*)$. Therefore $\mathrm{EH}(\gamma_*)$ is unbounded.

The volume is infinite as well. For an arbitrary point y, the left-hand side of the osculating inequality (9) at $\varepsilon = 2/3$ is $n(s) \cdot y + 3y_3$. Since $\|n(s)\| = 3\sqrt{5}$ and $|y_3| \le \|y\|$,

$$n(s) \cdot y + 3y_3 \le \big(3\sqrt{5} + 3\big)\|y\| = 3(\sqrt{5}+1)\,\|y\|.$$

The right-hand side is at most $9/2$ as soon as

$$\|y\| \le \rho_0, \qquad \rho_0 = \frac{3(\sqrt{5}-1)}{8},$$

because

$$3(\sqrt{5}+1) \cdot \frac{3(\sqrt{5}-1)}{8} = \frac{9}{2}.$$

Thus the closed ball $\overline{B}(0, \rho_0)$ is contained in $\mathrm{EH}(\gamma_*)$.

Fix $T > 0$ and choose $a = 2T/(3+2T) \in (0,1)$, so that $3a/(2(1-a)) = T$. The point $q_T = (0, 0, -T, 0)$ lies in $\mathrm{EH}(\gamma_*)$, and so do the four points of norm ρ_0

$$q_1 = (\rho_0, 0, 0, 0), \qquad q_2 = (0, \rho_0, 0, 0), \qquad q_3 = (0, 0, 0, \rho_0), \qquad q_4 = (0, 0, \rho_0, 0).$$

Their convex hull lies in $\mathrm{EH}(\gamma_*)$. The standard 4-simplex

$$\Delta = \big\{(u_1, u_2, u_3, u_4) : u_i \ge 0,\ u_1 + u_2 + u_3 + u_4 \le 1\big\}$$

has volume $1/24$: the iterated integral is

$$\int_0^1 \int_0^{1-u_1} \int_0^{1-u_1-u_2} \int_0^{1-u_1-u_2-u_3} du_4\, du_3\, du_2\, du_1 = \int_0^1 \frac{(1-u_1)^3}{6}\, du_1 = \frac{1}{24}.$$

The simplex with vertices q_T, q_1, q_2, q_3, q_4 is the image of Δ under the affine map $u \mapsto q_T + \sum_{j=1}^4 u_j(q_j - q_T)$. Lebesgue measure is multiplied by the absolute value of the determinant of a linear map, so the volume of the simplex is that absolute value divided by 24. The matrix with those columns is

$$\begin{pmatrix} \rho_0 & 0 & 0 & 0 \\ 0 & \rho_0 & 0 & 0 \\ T & T & T & T+\rho_0 \\ 0 & 0 & \rho_0 & 0 \end{pmatrix}.$$

The bottom row has a single nonzero entry, in column 3. Its cofactor sign is $(-1)^{4+3} = -1$, and the complementary minor

$$\begin{pmatrix} \rho_0 & 0 & 0 \\ 0 & \rho_0 & 0 \\ T & T & T+\rho_0 \end{pmatrix}$$

has determinant $\rho_0 \cdot \rho_0 \cdot (T + \rho_0)$. The full determinant equals $-\rho_0^3(T + \rho_0)$, so the simplex has volume

$$\frac{\rho_0^3(T + \rho_0)}{24}.$$

This quantity tends to infinity with T. A set containing subsets of arbitrarily large finite volume has infinite measure.

Lemma 10, applied with the factor $c = 1/\operatorname{length}(\gamma_*)$, produces a closed convex curve of length 1 whose elliptic hull is the corresponding homothety of $\mathrm{EH}(\gamma_*)$. That homothety is still unbounded and still of infinite volume. □

The mechanism is the position of p_*. On $\mathrm{EH}(e)$ the coordinate x_3 reaches $-3/2$. At $\varepsilon = 2/3$ the denominator $1 + \varepsilon x_3$ vanishes at p_* and is positive at every earlier point of the segment from the origin to p_*. Along the curve one has $e_3 \geq -1/2$, so the same denominator is at least $2/3$ and γ_* remains smooth. Points of the segment approaching p_* are sent to infinity, and Proposition 9 puts their images in $\mathrm{EH}(\gamma_*)$.

6 Compact hulls of arbitrarily large volume

Theorem 11 uses the endpoint $\varepsilon = 2/3$, where the hull ceases to be compact. For every smaller value, Proposition 9 keeps the hull compact. The compactness is not uniform in ε, and neither is the volume after the length has been normalized.

Lemma 12. *Let $r = 3/(2\sqrt{5})$. For each $\ell \in [0, 6/5]$, every point with third coordinate $-3/2 + \ell$ whose Euclidean distance to the x_3-axis is at most $\ell/2$ lies in $\operatorname{conv}\big(\overline{B}(0, r), p_*\big)$, hence in $\mathrm{EH}(e)$.*

Proof. The ball and the point p_* lie in $\mathrm{EH}(e)$ by Proposition 5, so their convex hull does as well. The case $\ell = 0$ is the single point p_*. Assume $0 < \ell \leq 6/5$.

The distance from the origin O to p_* is $3/2$. In the right triangle formed by O, p_* and a point of tangency T of a ray from p_* to the sphere, the right angle is at T. The leg OT has length r, so the cosine of the angle at O equals

$$\frac{r}{3/2} = \frac{1}{\sqrt{5}}.$$

The foot H of the perpendicular from T to the axis therefore lies between O and p_*, at distance

$$r \cdot \frac{1}{\sqrt{5}} = \frac{3}{2\sqrt{5}} \cdot \frac{1}{\sqrt{5}} = \frac{3}{10}$$

from the origin. The tangent hyperplane is $\{x_3 = -3/10\}$, and its distance from p_* is

$$\frac{3}{2} - \frac{3}{10} = \frac{6}{5}.$$

Its intersection with the ball is the set of points of the hyperplane at distance at most

$$\sqrt{r^2 - \Big(\frac{3}{10}\Big)^2}$$

from H. The quantity under the square root is

$$\frac{9}{20} - \frac{9}{100} = \frac{9}{25},$$

so the radius is $3/5$.

Consider a point q at axis-distance ℓ from p_* and at distance $d \le \ell/2$ from the axis. In coordinates, $p_* = (0, 0, -3/2, 0)$ and

$$q - p_* = (v_1, v_2, \ell, v_4), \qquad v_1^2 + v_2^2 + v_4^2 = d^2.$$

The ray $p_* + t(q - p_*)$ meets the tangent hyperplane when $-3/2 + t\ell = -3/10$, that is, at

$$t = \frac{6/5}{\ell} \ge 1.$$

The meeting point w is at distance $td \le 3/5$ from the axis, hence lies in the closed ball. Solving for q gives

$$q = \Big(1 - \frac{1}{t}\Big)p_* + \frac{1}{t}\, w,$$

a convex combination of p_* and a point of the ball. □

Lemma 13. *Let $\varepsilon \in [0, 2/3)$ and set $\eta = 1 - \frac{3}{2}\varepsilon \in (0, 1]$. Then $\mathrm{EH}(\gamma_\varepsilon)$ is compact,*

$$\operatorname{diam} \mathrm{EH}(\gamma_\varepsilon) \ge \frac{3}{2\eta},$$

and

$$\operatorname{Vol} \mathrm{EH}(\gamma_\varepsilon) \ge \frac{1}{384\sqrt{3}\,\eta}.$$

Proof. Compactness is the last statement of Proposition 9. The points 0 and p_* lie in $\mathrm{EH}(e) \subset D_\varepsilon$, and

$$\phi_\varepsilon(p_*) = \frac{p_*}{\eta}, \qquad \big\|\phi_\varepsilon(p_*) - \phi_\varepsilon(0)\big\| = \frac{3}{2\eta}.$$

Both images lie in $\mathrm{EH}(\gamma_\varepsilon)$, which is the diameter bound.

For the volume, fix $\ell \in [0, 1]$ and let $C(\ell)$ be the set of points whose third coordinate equals $-3/2 + \ell$ and whose first, second and fourth coordinates each have absolute value at most $\ell/(2\sqrt{3})$. A corner of this cube is at distance

$$\sqrt{3\Big(\frac{\ell}{2\sqrt{3}}\Big)^2} = \frac{\ell}{2}$$

from the axis, so the whole cube lies in the set described by Lemma 12. Thus $C(\ell) \subset \mathrm{EH}(e)$. The side length of the cube is $\ell/\sqrt{3}$, and its three-dimensional volume is

$$\Big(\frac{\ell}{\sqrt{3}}\Big)^3 = \frac{\ell^3}{3\sqrt{3}}.$$

On $C(\ell)$ the denominator of ϕ_ε is constant:

$$\Phi(\ell) = 1 + \varepsilon\Big(-\frac{3}{2} + \ell\Big) = \eta + \varepsilon\ell.$$

Since $\varepsilon < 2/3$, one has $\Phi(\ell) \ge \eta > 0$, so $C(\ell) \subset D_\varepsilon$. Proposition 9 puts $\phi_\varepsilon(C(\ell))$ in $\mathrm{EH}(\gamma_\varepsilon)$. On this slice, ϕ_ε multiplies every coordinate by $1/\Phi(\ell)$. The image lies in the hyperplane where the third coordinate equals

$$y_3(\ell) = \frac{-3/2 + \ell}{\Phi(\ell)},$$

and the cross-section in the remaining three coordinates is a cube of side $(\ell/\sqrt{3})/\Phi(\ell)$, hence of three-dimensional volume

$$\frac{\ell^3}{3\sqrt{3}\,\Phi(\ell)^3}.$$

Differentiate the height:

$$y_3'(\ell) = \frac{\Phi(\ell) - \varepsilon(-3/2+\ell)}{\Phi(\ell)^2} = \frac{1}{\Phi(\ell)^2} > 0,$$

where the numerator simplifies because $\Phi(\ell) = 1 + \varepsilon(-3/2+\ell)$. As ℓ runs from 0 to 1, the height y_3 is strictly increasing, so the image slices meet only along boundaries. Fubini's theorem in the coordinates (y_1, y_2, y_3, y_4) therefore gives

$$\mathrm{Vol}\big(\phi_\varepsilon(C)\big) = \int_0^1 \frac{\ell^3}{3\sqrt{3}\,\Phi(\ell)^3} \cdot \frac{1}{\Phi(\ell)^2}\, d\ell = \int_0^1 \frac{\ell^3}{3\sqrt{3}\,\Phi(\ell)^5}\, d\ell,$$

where $C = \bigcup_{\ell\in[0,1]} C(\ell)$. This set is contained in $\mathrm{EH}(\gamma_\varepsilon)$.

On $[0,1]$ one has $\varepsilon\ell \le \ell$, because $\varepsilon \le 2/3 \le 1$. Hence $\Phi(\ell) = \eta + \varepsilon\ell \le \eta + \ell$, and $\Phi^{-5} \ge (\eta+\ell)^{-5}$. The integral is at least

$$\frac{1}{3\sqrt{3}} \int_0^1 \frac{\ell^3}{(\eta+\ell)^5}\, d\ell.$$

The substitution $\ell = \eta u$ uses $\eta \le 1$, so the upper limit $1/\eta$ is at least 1:

$$\int_0^1 \frac{\ell^3}{(\eta+\ell)^5}\, d\ell = \eta^{-1} \int_0^{1/\eta} \frac{u^3}{(1+u)^5}\, du \ge \eta^{-1} \int_0^1 \frac{u^3}{(1+u)^5}\, du.$$

For $u \in [0,1]$ one has $1+u \le 2$, so $(1+u)^5 \le 32$ and $u^3/(1+u)^5 \ge u^3/32$. Therefore

$$\int_0^1 \frac{u^3}{(1+u)^5}\, du \ge \int_0^1 \frac{u^3}{32}\, du = \frac{1}{128},$$

and

$$\mathrm{Vol}\,\mathrm{EH}(\gamma_\varepsilon) \ge \frac{1}{3\sqrt{3}} \cdot \frac{1}{128\eta} = \frac{1}{384\sqrt{3}\,\eta}.$$

□

Theorem 14. *For every $M > 0$ there exists a closed convex curve in $\mathbb{R}^4$ of length 1 whose elliptic hull is compact and has both diameter and volume greater than M.*

Proof. Let $L_* = 3\pi\sqrt{37}/2$, the uniform length bound of Lemma 7. For $\eta \in (0,1]$ set $\varepsilon = (2/3)(1-\eta) \in [0, 2/3)$, and let γ_ε be the curve (7). Lemma 13 provides a compact elliptic hull of diameter at least $3/(2\eta)$ and volume at least $1/(384\sqrt{3}\,\eta)$, while Lemma 7 gives $0 < \mathrm{length}(\gamma_\varepsilon) \le L_*$.

Apply Lemma 10 with the factor $c = 1/\mathrm{length}(\gamma_\varepsilon)$. Then $c \ge 1/L_*$. The rescaled curve is closed and convex of length 1, and its elliptic hull is the homothety of factor c of a compact set, hence compact. Diameter scales by c and volume by c^4, so

$$\mathrm{diam} \ge \frac{1}{L_*} \cdot \frac{3}{2\eta} = \frac{3}{2\eta L_*},$$

$$\mathrm{Vol} \ge \frac{1}{L_*^4} \cdot \frac{1}{384\sqrt{3}\,\eta}.$$

Both quantities tend to infinity as $\eta \downarrow 0$. Any prescribed M is exceeded by choosing $\eta > 0$ small enough, for instance

$$\eta < \min\left(1,\ \frac{3}{2ML_*},\ \frac{1}{384\sqrt{3}\,L_*^4 M}\right)$$

when $M > 0$. The corresponding $\varepsilon = (2/3)(1-\eta)$ determines the curve explicitly. □

Corollary 15. *The normalized ellipse does not maximize elliptic-hull volume among closed convex curves of a given length. After normalizing length to* 1*, the quotient of e is finite, while Theorem 14 produces quotients larger than any finite number.*

Proof. Proposition 5 says that $\mathrm{EH}(e)$ is compact, so its volume is finite. Lemma 7 says that $\mathrm{length}(e) = 2\pi\sqrt{2} > 0$. The quotient of e is therefore finite. Homotheties multiply volume by the fourth power of the factor and length by the factor, so the quotient is scale-invariant. Theorem 14 produces length-one curves whose quotients exceed the quotient of e. □

Theorem 1 is the combination of Theorems 11 and 14. The first shows that the quotient may be infinite. The second shows that the quotient is already unbounded on the subclass of curves whose elliptic hull is compact, so the failure of (1) is not confined to a single degenerate parameter.

7 The ordinary convex hull

The same length bound does control the ordinary convex hull. The comparison makes precise what is special about the elliptic hull: the osculating inequalities see the point p_*, which lies outside the convex hull of the curve and is driven to infinity by the projective map, while every point of the convex hull remains at distance at most half the length from the curve.

Proposition 16. *Let γ be a closed rectifiable curve of length L in a Euclidean space. The diameter of the convex hull of its trace is at most $L/2$. Consequently there is a finite constant C_4, namely the volume of the Euclidean unit ball in $\mathbb{R}^4$, such that every closed curve of length L in $\mathbb{R}^4$ has convex hull of volume at most $C_4(L/2)^4$.*

Proof. Any two parameter values divide the domain circle into two arcs, and at least one of those arcs has length at most $L/2$. Integrating γ' along that arc,

$$\|\gamma(t_1) - \gamma(t_2)\| = \left\| \int \gamma'(t)\,dt \right\| \le \int \|\gamma'(t)\|\,dt \le \frac{L}{2}.$$

Thus the trace of γ has diameter at most $L/2$.

The convex hull has the same diameter. A point of the convex hull is a finite convex combination of points of the trace. If $x = \sum_i \lambda_i a_i$ and $y = \sum_j \mu_j b_j$ are two such combinations, then

$$x - y = \sum_{i,j} \lambda_i \mu_j (a_i - b_j),$$

so

$$\|x - y\| \le \sum_{i,j} \lambda_i \mu_j \|a_i - b_j\| \le \frac{L}{2}.$$

Fix a point p of the trace. Every point of the trace, and therefore every point of the convex hull, lies in the closed ball of radius $L/2$ about p. The volume of that ball is $C_4(L/2)^4$. □

No such constant exists for the elliptic hull. The curves of Theorems 11 and 14 all have length 1, while their elliptic volumes are infinite or arbitrarily large.

This is an AI-generated solution to a problem that was prompted by Paata Ivanisvili in Cursor with Grok 4.7.

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